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Practical Assignment 2 [100 marks] Deadline: 12th October 2025 at 22H00

1. Introduction

The final database assignment will be completed as a **group project**. This task is intended to give you practical experience in database design and implementation, while also allowing you to carry out some independent research. Your final submission will consist of a **written report** and a **working database system**, supported by queries and a Python/Java Script.

2. Project Tasks

Each group is required to:

- Develop a **business scenario or narrative** for a chosen problem domain
- Submit the scenario for **approval** before beginning further work.
- Identify and **extract business rules** from the scenario.
- Design an **Entity–Relationship Diagram (ERD)** that models the scenario.
- **Implement the ERD in MySQL**, including all necessary tables and constraints.
- Write and run a set of **SQL queries** against your database.
- Develop a **Python or Java program** that connects to your database and executes queries.
- Compile a **final report** containing the scenario, ERD, database implementation, queries, script, and your discussion of the work (**please see report template provided as a guideline**).

NB: The yellow-highlighted section has been finalized by now

3. Project Requirements

When designing your database:

- Your scenario must include **at least four entities**.
- There must be at least **one many-to-many relationship**, which you must decompose into a composite or bridge entity.
- Show an understanding of **weak and strong relationships**, and consider the possibility of multivariate entities.
- Discuss the **nature of attributes** in your report. This includes any derived attributes, multivalued attributes, or other special cases.
- Describe the **normalization steps** done in your scenario.
- The project should reflect both your **technical ability** and your **ability to apply database concepts thoughtfully**.

While most marks will be awarded for content, the **clarity and presentation** of your work will also be taken into account.

What is required to submit:

- **A Report in a PDF format with a GitHub link with the Python or Java code.** The GitHub Repository should be public for us to access it.

NB: Presentation: Each member of the group must indicate how/what they have contributed to the assignment in the report. Furthermore, a physical presentation would likely be required for final grading. The presentation dates would be communicated in due course.

SUBMISSION AND GRADING

Please submit the files named **TeamName_Practical_2.pdf**, which contains your report and the Java/Python implementation GitHub link. Also, please comment your teamName, student numbers, and file name in the Java/Python file(s).

Marks will be allocated as follows:

Task	Points
Question 1 - Script	20
Question 2 – Report	55
Question 3 - Presentation	25